

Patient:	Redacted	Legal Sex:	Male
D.O.B.:	Redacted	Family Doctor:	-
MRN:	-	Registration Date:	-
Account #:	-	Specimen #:	Redacted
Health Card #:	-		

Specimen Details

	Collected:	Received:	Type:	Source:
24C-101GMOO14	10/4/2024 1158	10/4/2024 1158	Blood	Blood, Venous

Hereditary Cancer Panel (Final result)

Referring Diagnosis:
Hereditary Breast/Ovarian/Prostate Cancer

Test Summary

Panel	Platform	Software	Genome Build
SOPHiA Custom Hereditary Cancer Solution (CHCS_AA_VI)	Illumina NextSeq 550	Illumina LRM v.2.4, SOPHiA DDM v.5.10	GRCh37

Interpretation

RESULT: No pathogenic germline variants were detected: However, a variant of uncertain significance (VUS) was detected in the PALB2 gene (see below for explanation)

Classification	Accession(Gene):cDNA,Protein	Zygosity
VUS	NM_024675.3(PALB2):c.2106A>G, p.(Ile702Met)	Heterozygous

Interpretation: No pathogenic variants were detected in any of the genes targeted by massively parallel sequencing and deletion/duplication analysis. Although this result decreases the likelihood of hereditary cancer; it does not exclude a diagnosis of a hereditary cancer syndrome

A VUS, c.2106A>G, p.(Ile702Met), was detected in the PALB2 gene. This missense variant is predicted to result in amino acid substitution of isoleucine at codon 702 with methionine. This variant is present at a low frequency in the general population in the Genome Aggregation Database (gnomAD ALL: 0.0011%, dbSNP: rs730881886; ACMGG PM2), as well as in unaffected individuals [PMID: 33471991]. Although the majority of pathogenic variants in PALB2 are truncating (ACMGG BP1), this variant has been reported in individuals with breast cancer or ovarian cancer [PMID: 35610400; 33471991; 26283626; 29522266] (ACMGG PS4_Supporting). This variant has also been reported in one individual with colorectal cancer [PMID: 33309985]. In silico analyses are concordant regarding the likely benign effect this variant may have on protein structure and function (ACMGG BP4), however; functional analyses have not been performed to verify these predictions. ClinVar contains an entry for this variant (Variation ID: 182764) where it is classified as uncertain by the majority of submitting laboratories. Given the available evidence, our laboratory currently classifies this a VUS (ACMGG: PS4_Supporting; PMZ, BPI, BPA)

Recommendations: Medical screening and management should rely on clinical findings and family history. An explanation of these results is available through consultation with the local Genetics clinic.

Please note: Variants determined to be benign or likely benign are not included on this report; but are available upon request.

HBOP1 Genes examined: ATM, BARD1, BRCA1, BRCA2, BRIP1, CDH1, CHEK2, EPCAM (3' copy number assessment only), HOXB13 (exon 1 only), MLH1, MSH2, MSH6, PALB2, PMS2, PTEN, RAD51C, RAD51D, STK11, TP53

Methodology: Massively parallel sequencing of coding exons and exon-intron junctions (+/-20 bp) of the above genes was performed with the Sophia CHCS_AA_v1 panel; sequence variant analysis and duplication/deletion analysis using sequence depth assessment was performed on the Sophia DDM software: Confirmation of copy number variants with an alternate method (MLPA or microarray) was performed when necessary and possible.

Variant Classification Rules: ACMG Guidelines. Richards et al, Genet Med. 17:405-424

Nomenclature Guidelines: HGVS Guidelines, varnomen.hgvs.org, Version 20.05.

Limitations: The following may not be detected: low-level mosaics (< 20% allele frequency), regulatory and deep intronic regions; breakpoints of large deletions or duplications. Diagnostic errors may occur due to rare sequence variants, inaccurate family histories or sample mixup.

Signed By

REDACTED on 10/04/24

Blood specimen REDACTED from Blood, Venous Unspecified. Ordered by REDACTED. Authorized by REDACTED, PhD FCCMG. Collected: 10/4/2024 1158, Received: 1158. Verified: 10/4/2024 1223. Resulted by Hedgefield Health Centre Molecular Genetics Lab.

Resulting Labs

Hedgefield General Hospital Laboratory - Molecular Genetics MOLGEN LAB, 123 Sample St.; Hedgefield A1B, 2C3

Lab Section: Genetics

**** END OF REPORT ****

If you have received this report in error, please return by Fax to Health Information Management at Hedgefield Health Centre at 123-456-7899. If you don't have access to fax, or have other questions, please call 123-456-7890 ext. 0001